

A clean-endpoint posterior sampler for cascaded flow priors

Abstract

We derive a posterior sampler for linear inverse problems under a cascaded pixel-domain flow prior. The image update is conditioned on the denoised estimate supplied by the prior rather than on the noisy interpolant. Three properties follow from this choice. First, the image marginal of the augmented target is the posterior itself, so the prior is not smoothed. Second, both conditionals of the augmented target are Gaussian, and both are therefore sampled exactly; the scheme uses no step size. Third, the one quantity that remains unspecified, the coupling covariance, admits an upper bound supplied by the model, and consequently need not be tuned. We state this bound, prove it under log-concavity of the prior, and show that it becomes asymptotically tight as the interpolant noise vanishes. We then propose a coupling covariance that is positive definite at every point of the schedule and coincides with the exact conditional covariance when the prior is Gaussian. The resulting algorithm requires one network evaluation and two linear solves per iteration, matching the cost of the samplers it replaces, and contains no continuous quantity that is searched over.

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1 Introduction

Let x be an image and $y = Ax + \eta\varepsilon$ a linear measurement with Gaussian noise. Given a pretrained generative prior, posterior sampling asks for draws from $p(x | y) \propto p(y | x)p(x)$ rather than for a single reconstruction. The difficulty is that the prior is available only through a learned score, and only at noise levels the model was trained on, so $p(x)$ itself can be neither evaluated nor differentiated at a clean image.

The standard remedy is variable splitting. An auxiliary variable is introduced, coupled to the image, and the two resulting conditionals are sampled in alternation. Applied to a diffusion or flow prior, this yields the plug-and-play and split-Gibbs samplers, in which the coupling is $\mathcal{N}(x; z, \rho^2 I)$ and ρ is a design parameter annealed toward zero. We treat a case in which the auxiliary variable is not free. For a cascaded flow model, the natural auxiliary is the interpolant of the model itself, the coupling operator is its degradation operator, and the coupling width is determined by the schedule.

Two constructions are available, and they differ in which factor of the joint density is kept intact. The first couples the image to the interpolant through the flat-prior Gaussian. The learned marginal then remains in the conditional for the auxiliary variable, which introduces both a step size and a smoothing of the prior. The second, developed here, couples through the true denoising conditional. In this factorization the learned marginal cancels, and the step size and the smoothing are removed together. The construction instead requires the covariance of the denoising conditional, which is not directly computable. Sections 4–6 establish how far that requirement can be met.

Contributions.

1. A factorization of the augmented posterior whose image marginal is exact (Section 3), so no smoothed prior is introduced.

2. An upper bound on the coupling covariance supplied by the model itself (Lemma 3), proved under log-concavity, together with an analysis showing the bound becomes asymptotically tight as the interpolant noise vanishes (Section 8.3).
3. A choice of coupling covariance that is positive definite at every τ including the degenerate boundary, exact when the prior is Gaussian, and free of tuned parameters (Section 4.2).
4. A single-solve estimator for the clean endpoint (Proposition 7), and a proof that both conditionals are sampled exactly (Section 6), giving an algorithm with no step size and no searched hyperparameter (Algorithm 1).

2 Setup

2.1 The cascaded flow model

The prior is a cascade of K stages over increasing resolutions. Stage k carries boundary coefficients $0 \leq s_k < e_k \leq 1$, with $s_0 = 0$ and $e_{K-1} = 1$, and a fixed degradation operator G , assumed symmetric with $0 \preceq G \preceq I$; in practice G is a blur, often diagonalized by the discrete Fourier transform. Within stage k a time $\tau \in [0, 1]$ interpolates linearly between the two stage boundaries. Writing $x_1^k \in \mathbb{R}^n$ for the clean image at this scale and $x_0^k \sim \mathcal{N}(0, I_n)$ for an independent noise endpoint, the interpolant is

$$\boxed{x_\tau^k = H_\tau^k x_1^k + \sigma_\tau^k x_0^k} \quad H_\tau^k := (1 - \tau)s_k G + \tau e_k I, \quad \sigma_\tau^k := (1 - \tau)(1 - s_k) + \tau(1 - e_k). \quad (1)$$

Both boundaries are recovered as special cases. Setting $\tau = 0$ gives $x_{\text{start}}^k = s_k G x_1^k + (1 - s_k)x_0^k$, and setting $\tau = 1$ gives $x_{\text{end}}^k = e_k x_1^k + (1 - e_k)x_0^k$. The operator H_τ^k is a polynomial in G and is therefore symmetric. It is invertible for $\tau > 0$. At $\tau = 0$ it reduces to $s_k G$, which is rank-deficient whenever G is. The width σ_τ^k is a positive scalar except at $\tau = 1$ of the final stage, where $e_{K-1} = 1$ and hence $\sigma_\tau^k = 0$.

Write p_1^k for the law of X_1^k and p_τ^k for the law of X_τ^k , the *flow marginal* at time τ of stage k . Neither is available in closed form. What the pretrained network supplies is a velocity field $v_\theta(x_\tau^k, \tau, k)$ approximating the displacement between the two boundaries,

$$d_\tau^k := x_{\text{end}}^k - x_{\text{start}}^k = B_k x_1^k - (e_k - s_k)x_0^k, \quad B_k := e_k I - s_k G, \quad (2)$$

with an error we model as $v_\theta = d_\tau^k + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \gamma^2 I)$ independent of X_0^k . The variance γ^2 is a property of the trained network. It is measured on held-out data by (20) below and tabulated per (k, τ) , and is therefore not a free parameter. Between stages the state is carried by an upsampling operator, $x_1^{k+1} = U^{(1)}(x_1^k)$ with a fresh $x_0^{k+1} \sim \mathcal{N}(0, I)$.

One further operator is used throughout. It is again a polynomial in G and consequently commutes with H_τ^k and B_k . It does not depend on τ .

$$N_k := e_k(1 - s_k)I - s_k(1 - e_k)G = \sigma_\tau^k B_k + (e_k - s_k)H_\tau^k, \quad (3)$$

the second equality being verified in Appendix A.1.

2.2 The inverse problem

At scale k the measurement operator is $A_k := A U^{K-1-k}$, the composition of the forward operator with the upsampling to full resolution, and

$$p(y | x_1^k) = \mathcal{N}(y; A_k x_1^k, \eta^2 I_m) \propto \exp\left(-\frac{1}{2\eta^2} \|A_k x_1^k - y\|^2\right), \quad (4)$$

with η the known measurement noise level. The object of interest is $p(x_1^k | y) \propto p(y | x_1^k) p_1^k(x_1^k)$. This density cannot be sampled directly, for two reasons. First, p_1^k is often supported on a set of measure zero, and hence its score does not exist away from that set. Second, the network represents p_τ^k rather than p_1^k , so the required score is unavailable in any case.

2.3 Notation

Table 1 collects the symbols used below. Throughout, $\preceq$ denotes the Loewner order and $\text{Cov}(\cdot | \cdot)$ a conditional covariance. A hat denotes a computed estimate, as distinct from a state variable or an exact conditional mean.

Table 1: Symbols.

x_1^k, x_0^k	clean image and noise endpoint at stage k ; the two state variables
x_τ^k	interpolant (1); a deterministic function of the pair
$\bar{x}_1^k, \bar{x}_0^k$	conditional means $\mathbb{E}[X_1^k x_\tau^k], \mathbb{E}[X_0^k x_\tau^k]$
$\hat{x}_1^k$	computed estimate of $\bar{x}_1^k$, from one network evaluation
G, H_τ^k, B_k, N_k	degradation, interpolant operator, displacement operator, elimination operator
$\sigma_\tau^k, \gamma^2, \eta^2$	interpolant width, velocity-error variance, measurement-noise variance
C, S	coupling covariance, prior covariance surrogate
M_τ^{den}	precision of the clean-image conditional, (21)

3 The target and its two conditionals

3.1 Factorizing through the denoising conditional

The pair (X_1^k, X_τ^k) has a joint density, and it may be factorized in either direction. The choice made here is

$$\pi(x_1^k, x_\tau^k | y) \propto \underbrace{p(y | x_1^k)}_{\text{likelihood (4)}} \underbrace{p(x_1^k | x_\tau^k)}_{\text{denoising conditional}} \underbrace{p_\tau^k(x_\tau^k)}_{\text{flow marginal}}, \quad (5)$$

in which y depends on x_τ^k only through x_1^k , since (4) involves x_1^k alone.

Proposition 1 (The image marginal is exact). *Marginalizing x_τ^k out of (5) returns $\pi(x_1^k | y) \propto p(y | x_1^k) p_1^k(x_1^k)$, the posterior of Section 2.*

Proof. The product of the last two factors of (5) is $p(x_1^k | x_\tau^k) p_\tau^k(x_\tau^k)$, which is by definition the joint density of (X_1^k, X_τ^k) . Its x_1^k -marginal is $p_1^k(x_1^k)$, and $p(y | x_1^k)$ does not depend on x_τ^k , so it passes through the integral. $\square$

Proposition 1 motivates the factorization. The augmentation is exact. Consequently no smoothing of the prior is introduced, and no annealing parameter is required to remove one.

3.2 The two conditionals

Reading the conditionals off (5) — keep every factor containing the variable of interest, discard the rest, since such a factor is a constant absorbed into the normalization — gives a data step and a renoising step.

Proposition 2 (The two conditionals). *For (5),*

$$\pi(x_1^k | x_\tau^k, y) \propto p(y | x_1^k) p(x_1^k | x_\tau^k), \quad (6)$$

$$\pi(x_\tau^k | x_1^k, y) = p(x_\tau^k | x_1^k) = \mathcal{N}(x_\tau^k; H_\tau^k x_1^k, (\sigma_\tau^k)^2 I). \quad (7)$$

Proof. For (6), hold x_τ^k fixed; the factor $p_\tau^k(x_\tau^k)$ is then a constant. For (7), hold x_1^k fixed; the likelihood is a constant, and the remaining product $p(x_1^k | x_\tau^k) p_\tau^k(x_\tau^k)$ is the joint density of the pair, whose x_τ^k -conditional is $p(x_\tau^k | x_1^k)$. By (1) that conditional is the law of $H_\tau^k x_1^k + \sigma_\tau^k X_0^k$ with $X_0^k \sim \mathcal{N}(0, I_n)$, which is the stated Gaussian. $\square$

We draw out two consequences, on which the remainder of the note depends.

Cancellation of the learned marginal. The factor p_τ^k does not appear in (7), having cancelled against the p_τ^k implicit in $p(x_1^k | x_\tau^k)$. This cancellation is specific to the factorization (5). If the image is instead coupled to the interpolant through the flat-prior Gaussian $\mathcal{N}(x_\tau^k; H_\tau^k x_1^k, (\sigma_\tau^k)^2 I)$, then $p_\tau^k(x_\tau^k)$ remains in the x_τ^k -conditional. Because that factor is intractable, a Langevin step is then required. The step size and the prior smoothing therefore arise from the same source, and neither is present here.

Relocation of the obstruction. The obstruction is relocated rather than removed. By Bayes' rule, $p(x_1^k | x_\tau^k) \propto p_1^k(x_1^k) \mathcal{N}(x_\tau^k; H_\tau^k x_1^k, (\sigma_\tau^k)^2 I)$, which contains the unavailable p_1^k . The remaining difficulty is thus confined to that single factor, and Section 4 addresses it.

3.3 The Gaussian surrogate, and the two gradients

We approximate the denoising conditional by a Gaussian centred at its own mean,

$$p(x_1^k | x_\tau^k) \approx \mathcal{N}(x_1^k; \hat{x}_1^k, C), \quad \hat{x}_1^k \approx \mathbb{E}[X_1^k | x_\tau^k], \quad C \approx \text{Cov}(X_1^k | x_\tau^k). \quad (8)$$

Equation (8) is the only approximation in the construction. Section 5 computes $\hat{x}_1^k$ and Section 4 selects C . Differentiating (6)–(7) with (8) in place gives the two gradients,

$$\boxed{\nabla_{x_1^k} \log \pi(x_1^k | x_\tau^k, y) = \underbrace{\frac{1}{\eta^2} A_k^\top (y - A_k x_1^k)}_{\text{likelihood}} - \underbrace{C^{-1} (x_1^k - \hat{x}_1^k)}_{\text{pull toward the denoised estimate}}} \quad (9)$$

$$\boxed{\nabla_{x_\tau^k} \log \pi(x_\tau^k | x_1^k, y) = -\frac{1}{(\sigma_\tau^k)^2} (x_\tau^k - H_\tau^k x_1^k)} \quad (10)$$

Both gradients are affine in the variable being updated. Section 6 shows that affineness is the condition under which a conditional may be sampled without a step size. Both blocks therefore admit exact draws.

4 The coupling covariance

The matrix C scales the entire contribution of the prior to every image update, and therefore cannot be selected arbitrarily. This section establishes an upper bound on C supplied by the model, and proposes a choice within that bound. The proposed choice is positive definite at every τ and coincides with the exact conditional covariance when the prior is Gaussian.

4.1 The bound

We first state the bound informally. The interpolant alone determines x_1^k only up to the flat-prior deconvolution, whose covariance is $(\sigma_\tau^k)^2(H_\tau^k)^{-2}$. A denoised estimate should be at least as informative as that deconvolution. Under one hypothesis this argument can be made exact.

Lemma 3 (Bound on the coupling covariance). *Let $\tau > 0$ and suppose p_1^k is log-concave. Then for every x_τ^k ,*

$$\boxed{\text{Cov}(X_1^k | x_\tau^k) \preceq (\sigma_\tau^k)^2 (H_\tau^k)^{-2}} \iff \text{Cov}(X_1^k | x_\tau^k)^{-1} \succeq \frac{1}{(\sigma_\tau^k)^2} (H_\tau^k)^\top H_\tau^k, \quad (11)$$

with equality when p_1^k is flat.

Proof. Write $p(x_1^k | x_\tau^k) \propto e^{-V(x_1^k)}$ with $V(x_1^k) = -\log p_1^k(x_1^k) + \|x_\tau^k - H_\tau^k x_1^k\|^2 / 2(\sigma_\tau^k)^2$. Differentiating twice in x_1^k , the second term contributes the constant $(H_\tau^k)^\top H_\tau^k / (\sigma_\tau^k)^2$ and the first is positive semidefinite because p_1^k is log-concave, so $\nabla^2 V \succeq \Lambda := (H_\tau^k)^\top H_\tau^k / (\sigma_\tau^k)^2$, which is positive definite for $\tau > 0$. The Brascamp–Lieb variance inequality states that a density $\propto e^{-V}$ with $\nabla^2 V \succeq \Lambda \succ 0$ has covariance at most Λ^{-1} . Hence $\text{Cov}(X_1^k | x_\tau^k) \preceq (\sigma_\tau^k)^2 (H_\tau^k)^{-2}$, using symmetry of H_τ^k . If p_1^k is flat the first term of V vanishes and $p(x_1^k | x_\tau^k) = \mathcal{N}((H_\tau^k)^{-1} x_\tau^k, (\sigma_\tau^k)^2 (H_\tau^k)^{-2})$ exactly. $\square$

The right-hand form of (11) is a lower bound on the precision C^{-1} , and it is the precision that enters the algorithm. Any admissible C therefore yields an image update at least as sharp as the update the interpolant alone would supply.

4.2 Choosing C

Saturating (11) by setting $C = (\sigma_\tau^k)^2 (H_\tau^k)^{-2}$ is the most conservative admissible choice, and it removes C from the implementation. This choice has one defect. Since H_τ^k is rank-deficient at $\tau = 0$, the saturated C is undefined there, and the precision $(H_\tau^k)^\top H_\tau^k / (\sigma_\tau^k)^2$ is singular on $\ker(G)$. A ridge is consequently required at the start of every stage.

Rather than adding an arbitrary ridge, we use the information the problem already supplies. The bound (11) uses no property of the prior beyond log-concavity. Suppose a covariance surrogate S is available, with $\text{Cov}(X_1^k) \preceq S$. Combining the two sources of information then gives

$$\boxed{C^{-1} := \frac{1}{(\sigma_\tau^k)^2} (H_\tau^k)^\top H_\tau^k + S^{-1}} \quad (12)$$

which is the choice adopted here. Three properties justify it, and Proposition 4 states them. In practice $S = s^2 I$ suffices, where s^2 is the per-pixel variance of the training images at scale k . This quantity is measured from the data rather than searched over.

Proposition 4 (Properties of (12)). *Let $S \succ 0$ and let C be given by (12). Then*

- (a) $C^{-1} \succeq S^{-1} \succ 0$ for every $\tau \in [0, 1]$, including $\tau = 0$; no ridge is needed anywhere;
- (b) $C \preceq (\sigma_\tau^k)^2 (H_\tau^k)^{-2}$, so (12) is admissible under Lemma 3, and strictly tighter than the saturated choice;
- (c) if $p_1^k = \mathcal{N}(0, S)$ then $C = \text{Cov}(X_1^k | x_\tau^k)$ exactly.

Proof. (a) $(H_\tau^k)^\top H_\tau^k \succeq 0$ always, so $C^{-1} \succeq S^{-1} \succ 0$; at $\tau = 0$ the first term is $s_k^2 G^2 / (\sigma_\tau^k)^2$, which is merely singular rather than negative, so the sum stays positive definite. (b) Adding $S^{-1} \succeq 0$ to the precision can only decrease the covariance in the Loewner order, and (11) is

the covariance corresponding to the first term alone. (c) For a Gaussian prior, $p(x_1^k | x_\tau^k) \propto \mathcal{N}(x_1^k; 0, S) \mathcal{N}(x_\tau^k; H_\tau^k x_1^k, (\sigma_\tau^k)^2 I)$ is Gaussian with precision $S^{-1} + (H_\tau^k)^\top H_\tau^k / (\sigma_\tau^k)^2$, which is (12). $\square$

Part (c) is the strongest of the three. By (c), (12) is not a ridge introduced for conditioning, but the exact conditional covariance when the prior is Gaussian; by (b), it remains admissible for any log-concave prior. Part (a) removes the boundary case at $\tau = 0$. Consequently, and in contrast to the interpolant-coupled construction, this route requires neither a properness argument nor a ridge parameter.

Choosing S . Two prescriptions are available, and Sections 8.2–8.3 analyse both. The first uses image statistics. Take S diagonal in the same basis as G ; when G is diagonalized by the discrete Fourier transform, so is H_τ^k , and setting

$$S(\omega) = \mathbb{E} |\hat{x}_1^k(\omega)|^2, \quad (13)$$

the average power spectrum of the training images at scale k , makes C^{-1} diagonal in that basis as well, so the factor $S^{-1/2}$ in (22) is a pointwise multiplication. The isotropic reduction $S = s^2 I$ uses the per-pixel variance in place of the spectrum.

The second prescription uses no image statistics at all. Setting $s^2 = c^2 (\sigma_\tau^k)^2$ for a dimensionless constant c gives

$$C^{-1} = \frac{1}{(\sigma_\tau^k)^2} \left[(H_\tau^k)^\top H_\tau^k + c^{-2} I \right], \quad (14)$$

that is, a constant ridge c^{-2} applied to $(H_\tau^k)^\top H_\tau^k$. This choice is positive definite at every τ and admissible under Lemma 3, and it requires only the single constant c . We note that it has the wrong dependence on τ : comparing (14) with (12) shows that the exact ridge is $(\sigma_\tau^k)^2 / s^2$, which decreases as σ_τ^k does, whereas c^{-2} is constant. Section 8.2 quantifies the resulting error and shows it to be bounded.

Contribution of the two terms along the schedule. The two terms of (12) trade off as τ advances, and the dependence is the favourable one. At $\tau = 0$ the first term is $s_k^2 G^2 / (\sigma_\tau^k)^2$, which vanishes on $\ker(G)$, so the prior surrogate supplies the entire precision in exactly the directions the interpolant does not constrain. As $\tau \rightarrow 1$ the first term grows like $e_k^2 / (\sigma_\tau^k)^2$ while S^{-1} remains fixed, so the surrogate’s relative contribution decreases monotonically. The surrogate therefore has greatest influence where the interpolant is least informative, and least influence where it is most informative. We note that the ridge is required at large σ_τ^k rather than at small σ_τ^k , since the degeneracy arises from H_τ^k and not from σ_τ^k .

5 Evaluating the clean endpoint

It remains to compute $\hat{x}_1^k$. The network reports a displacement rather than an endpoint, so the endpoint must be recovered from it. The recovery is an exact elimination rather than an estimate, because the same noise realization appears at both stage boundaries.

5.1 Two equations, two unknowns

Proposition 5 (The conditional-mean relations). *Write $\bar{x}_1^k = \mathbb{E}[X_1^k \mid x_\tau^k]$ and $\bar{x}_0^k = \mathbb{E}[X_0^k \mid x_\tau^k]$. Then*

$$\underbrace{x_\tau^k}_{\text{known}} = H_\tau^k \bar{x}_1^k + \sigma_\tau^k \bar{x}_0^k, \quad (15)$$

$$\underbrace{d_\tau^k}_{\text{estimated by } v_\theta} = B_k \bar{x}_1^k - (e_k - s_k) \bar{x}_0^k. \quad (16)$$

Proof. Both (1) and (2) are linear in (x_1^k, x_0^k) . Take conditional expectations given $X_\tau^k = x_\tau^k$ and use linearity; on the left of (15) the conditioning variable is known, so $\mathbb{E}[X_\tau^k \mid x_\tau^k] = x_\tau^k$. $\square$

Equations (15)–(16) are two vector equations in the two unknown conditional means, so either mean may be eliminated exactly. This is a consequence of the shared noise endpoint. It also explains why a least-squares fit treating the two boundaries as independently corrupted observations is inappropriate here: the two residuals are collinear, and there is therefore no independent error to average down.

Proposition 6 (The clean endpoint, exactly). *Eliminating $\bar{x}_0^k$ from (15)–(16) gives*

$$N_k \bar{x}_1^k = (e_k - s_k) x_\tau^k + \sigma_\tau^k d_\tau^k. \quad (17)$$

Proof. Multiply (15) by $(e_k - s_k)$ and (16) by σ_τ^k and add. The $\bar{x}_0^k$ terms are $\sigma_\tau^k(e_k - s_k)\bar{x}_0^k$ and $-\sigma_\tau^k(e_k - s_k)\bar{x}_0^k$ and cancel. The bracket multiplying $\bar{x}_1^k$ is $(e_k - s_k)H_\tau^k + \sigma_\tau^k B_k$, which is N_k by (3); the operators commute, being polynomials in G , so the order of multiplication is immaterial. $\square$

5.2 Regularizing for the velocity error

Substituting v_θ for d_τ^k in (17) introduces an error $\sigma_\tau^k \varepsilon$ into the right-hand side. Since N_k may in addition be ill-conditioned, the solve must be regularized. The regularizer is determined rather than chosen, because the model specifies $X_0^k \sim \mathcal{N}(0, I)$ exactly.

Proposition 7 (The regularized endpoint, in one solve). *Model $v_\theta = d_\tau^k + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \gamma^2 I)$ independent of X_0^k , and let $\hat{x}_0^k$ be the posterior mean of X_0^k under (15)–(16) with that error model and the prior $\mathcal{N}(0, I)$, namely the solution of*

$$[N_k^2 + \gamma^2 (H_\tau^k)^2] \hat{x}_0^k = N_k (B_k x_\tau^k - H_\tau^k v_\theta). \quad (18)$$

Set $\hat{x}_1^k := (H_\tau^k)^{-1} (x_\tau^k - \sigma_\tau^k \hat{x}_0^k)$. Then, for $\tau > 0$,

$$\boxed{[N_k^2 + \gamma^2 (H_\tau^k)^2] \hat{x}_1^k = N_k [(e_k - s_k) x_\tau^k + \sigma_\tau^k v_\theta] + \gamma^2 H_\tau^k x_\tau^k} \quad (19)$$

— one solve, against the same operator as (18).

Proof. Appendix A.2. $\square$

We make three remarks. First, the regularizer in (18) is $\gamma^2 (H_\tau^k)^2$, derived from a prior that holds by construction rather than assumed about images. Equation (19) is not an independently regularized fit of x_1^k , but a rewriting of that solution, and it therefore inherits the same derivation. Second, as $\gamma \rightarrow 0$, (19) reduces to (17). Third, as $\gamma \rightarrow \infty$ it reduces to $\hat{x}_1^k = (H_\tau^k)^{-1} x_\tau^k$. That is, when the velocity estimate is uninformative, the estimator returns the flat-prior deconvolution of the interpolant, which is the centre used by the interpolant-coupled sampler. The two constructions consequently coincide in the limit of an uninformative network, and differ by an amount that grows with the information the network supplies.

Measuring γ^2 . On held-out data both endpoints are known, so the exact displacement $d_\tau^k = B_k x_1^k - (e_k - s_k) x_0^k$ of (2) can be formed and

$$\gamma^2 \approx \frac{1}{n} \mathbb{E} \|v_\theta(x_\tau^k, \tau, k) - d_\tau^k\|^2, \quad (20)$$

tabulated per (k, τ) and stored with the model. Equation (20) is a measurement rather than a search. If the velocity error is found to be coloured rather than white, then $\gamma^2 I$ generalizes to a covariance Σ_ε and (18) becomes $[N_k(H_\tau^k \Sigma_\varepsilon H_\tau^k)^{-1} N_k + I] \hat{x}_0^k = N_k(H_\tau^k \Sigma_\varepsilon H_\tau^k)^{-1} (B_k x_\tau^k - H_\tau^k v_\theta)$, still one measurement per frequency band.

6 Both conditionals are sampled exactly

The claim that the scheme uses no step size rests on two elementary facts. The first gives a condition under which a density is Gaussian. The second gives a method for drawing from such a density at the cost of one linear solve.

Lemma 8 (An affine score means a Gaussian). *Let π be a probability density on $\mathbb{R}^n$ with $\nabla \log \pi(x) = b - Mx$ for all x , where M is symmetric positive definite. Then $\pi = \mathcal{N}(M^{-1}b, M^{-1})$.*

Proof. The field $x \mapsto b - Mx$ is the gradient of $\phi(x) = b^\top x - \frac{1}{2} x^\top Mx$, and on $\mathbb{R}^n$ a function is determined by its gradient up to an additive constant, so $\log \pi = \phi + \text{const}$. Completing the square, $\phi(x) = -\frac{1}{2} (x - M^{-1}b)^\top M (x - M^{-1}b) + \text{const}$, so $\pi(x) \propto \exp(-\frac{1}{2} (x - M^{-1}b)^\top M (x - M^{-1}b))$, integrable because $M \succ 0$. $\square$

Lemma 9 (Exact draw by one linear solve). *Let $M = \sum_j R_j^\top R_j \succ 0$, each R_j available only as a matrix-vector product, and let $b \in \mathbb{R}^n$. Draw $\xi_j \sim \mathcal{N}(0, I)$ independently and set $\zeta = \sum_j R_j^\top \xi_j$. Then $\text{Cov}(\zeta) = M$, and the solution x of $Mx = b + \zeta$ has law $\mathcal{N}(M^{-1}b, M^{-1})$. Neither M , nor M^{-1} , nor $M^{1/2}$ need be formed.*

Proof. $\text{Cov}(\zeta) = \sum_j R_j^\top \text{Cov}(\xi_j) R_j = \sum_j R_j^\top R_j = M$ by independence. Then $x = M^{-1}b + M^{-1}\zeta$ is an affine image of a Gaussian, hence Gaussian, with mean $M^{-1}b$ and covariance $M^{-1}MM^{-1} = M^{-1}$. Only the products $R_j v$, $R_j^\top w$ and the action of M are used, so the system may be solved by conjugate gradients. $\square$

6.1 Block 1: the image

Equation (9) is of the form $b - Mx$ with

$$M_\tau^{\text{den}} := \frac{1}{\eta^2} A_k^\top A_k + C^{-1} = \frac{1}{\eta^2} A_k^\top A_k + \frac{1}{(\sigma_\tau^k)^2} (H_\tau^k)^\top H_\tau^k + S^{-1}, \quad b = \frac{1}{\eta^2} A_k^\top y + C^{-1} \hat{x}_1^k, \quad (21)$$

using (12). By Proposition 4(a), $M_\tau^{\text{den}} \succ 0$ for every τ , so Lemma 8 makes the conditional Gaussian and Lemma 9 applies with the three factors $R_1 = A_k/\eta$, $R_2 = H_\tau^k/\sigma_\tau^k$ and $R_3 = S^{-1/2}$. The update is

$\text{Block 1: } M_\tau^{\text{den}} x_1^k = \frac{1}{\eta^2} A_k^\top y + C^{-1} \hat{x}_1^k + \frac{1}{\eta} A_k^\top \xi_y + \frac{1}{\sigma_\tau^k} (H_\tau^k)^\top \xi_h + S^{-1/2} \xi_s$

(22)

with $\xi_y \sim \mathcal{N}(0, I_m)$ and $\xi_h, \xi_s \sim \mathcal{N}(0, I_n)$ independent. With $S = s^2 I$ the third factor is ξ_s/s , so no square root is computed.

6.2 Block 2: the interpolant

Equation (10) is $b - Mx$ with $M = I/(\sigma_\tau^k)^2$ and $b = H_\tau^k x_1^k/(\sigma_\tau^k)^2$, so by Lemma 8 the conditional is $\mathcal{N}(H_\tau^k x_1^k, (\sigma_\tau^k)^2 I)$ — as Proposition 2 already states. Here M is a multiple of the identity, so Lemma 9 needs no solve at all and the update is a direct draw:

$$\boxed{\text{Block 2: } x_\tau^k \leftarrow H_\tau^k x_1^k + \sigma_\tau^k \xi_0, \quad \xi_0 \sim \mathcal{N}(0, I_n)} \quad (23)$$

that is, a fresh renoising of the current image at the schedule’s own noise level.

6.3 Absence of a step size

A step size is required when a conditional is non-Gaussian, and the non-Gaussianity enters through the learned marginal p_τ^k . Proposition 2 showed that this factor cancels from (7). It is absent from (6) by construction, since x_τ^k is held fixed there. Because p_τ^k appears in neither conditional, Lemma 8 applies to both, and both draws are exact.

We state the comparison with a Langevin scheme precisely, since the difference is not one of cost alone. For a Gaussian target of covariance Σ , a preconditioned Langevin step with $P = \Sigma$ moves the drift fully to the mean only at step size $h = 2$, and at that step size the injected noise has covariance 2Σ . Separately, the unpreconditioned step on (10) has stationary variance $2(\sigma_\tau^k)^2/(2 - h/2(\sigma_\tau^k)^2)$, which equals $(\sigma_\tau^k)^2$ only in the limit $h \rightarrow 0$, where the chain does not move. The drift therefore requires $h = 2$ and the noise requires $h = 1$. No step size reproduces an exact draw. Consequently (22)–(23) are not a limiting case of a Langevin scheme.

7 The algorithm

Collecting (19), (22) and (23) gives Algorithm 1. The cost per inner iteration is one network evaluation and two linear solves. The first, (19), is a polynomial in G alone: it degenerates to the scalar $e_k - s_k$ when $G = I$, and reduces to a pointwise division and two transforms whenever G is diagonalized by the discrete Fourier transform. The second, (22), mixes $A_k^\top A_k$ with the other two terms and requires conjugate gradients.

Algorithm 1 The clean-endpoint sampler. Both blocks are exact draws, so there is no step size anywhere and no continuous quantity is searched over: γ^2 is measured by (20) and S is the prior covariance surrogate of (12). By Proposition 4(a) the operator M_τ^{den} is positive definite at every τ , so no ridge is required at $\tau = 0$.

Require: y, A, η , stages K , time grid $\{\tau\}$, inner iterations S_{it} , prior covariance S , velocity-error variance $\gamma^2(k, \tau)$, CG iterations L

```

1:  $x_1^0 \leftarrow 0$ 
2: for  $k = 0, \dots, K - 1$  do
3:    $x_0^k \sim \mathcal{N}(0, I_n)$  ▷ fresh noise endpoint for the stage
4:   for  $\tau$  on the stage grid do
5:      $H_\tau^k \leftarrow (1 - \tau)s_k G + \tau e_k I$ ;  $\sigma_\tau^k \leftarrow (1 - \tau)(1 - s_k) + \tau(1 - e_k)$ 
6:      $N_k \leftarrow e_k(1 - s_k)I - s_k(1 - e_k)G$  ▷ (3); independent of  $\tau$ 
7:      $C^{-1} \leftarrow \frac{1}{(\sigma_\tau^k)^2} (H_\tau^k)^\top H_\tau^k + S^{-1}$ ;  $M_\tau^{\text{den}} \leftarrow \frac{1}{\eta^2} A_k^\top A_k + C^{-1}$  ▷ (12), (21)
8:      $x_\tau^k \leftarrow H_\tau^k x_1^k + \sigma_\tau^k x_0^k$  ▷ enter  $(x_1^k, x_\tau^k)$  coordinates at this  $\tau$ 
9:     for  $s = 1, \dots, S_{\text{it}}$  do
       Endpoint — one solve, Prop. 7
10:       $v \leftarrow v_\theta(x_\tau^k, \tau, k)$  ▷ one network evaluation
11:      solve  $[N_k^2 + \gamma^2(H_\tau^k)^2] \hat{x}_1^k = N_k[(e_k - s_k)x_\tau^k + \sigma_\tau^k v] + \gamma^2 H_\tau^k x_\tau^k$  ▷ (19)
       Block 1: exact draw of  $x_1^k$  from  $\pi(x_1^k | x_\tau^k, y)$ 
12:       $\xi_y \sim \mathcal{N}(0, I_m)$ ;  $\xi_h, \xi_s \sim \mathcal{N}(0, I_n)$ 
13:      solve  $M_\tau^{\text{den}} x_1^k = \frac{1}{\eta^2} A_k^\top y + C^{-1} \hat{x}_1^k + \frac{1}{\eta} A_k^\top \xi_y + \frac{1}{\sigma_\tau^k} (H_\tau^k)^\top \xi_h + S^{-1/2} \xi_s$  ▷ (22); Lem. 9
       Block 2: exact draw of  $x_\tau^k$  from  $\pi(x_\tau^k | x_1^k, y)$  (no solve)
14:       $\xi_0 \sim \mathcal{N}(0, I_n)$ ;  $x_\tau^k \leftarrow H_\tau^k x_1^k + \sigma_\tau^k \xi_0$  ▷ (23)
15:    end for end inner iterations
16:     $x_0^k \leftarrow (x_\tau^k - H_\tau^k x_1^k) / \sigma_\tau^k$  ▷ leave the coordinates;  $x_0^k$  carries to the next  $\tau$ 
17:  end for end time grid  $\tau$  (stage  $k$  complete)
18:   $x_1^{k+1} \leftarrow U^{(1)}(x_1^k)$ ;  $x_0^{k+1} \sim \mathcal{N}(0, I)$  ▷ ancestral transition to the next scale
19: end for end scales  $k$ 
20: return  $x_1^{K-1}$ 

```

Choice of state variable between grid points. The inner loop operates in (x_1^k, x_τ^k) , but the state carried across the time grid is x_0^k . The law of X_0^k is $\mathcal{N}(0, I_n)$ at every τ , whereas the law of X_τ^k is p_τ^k and varies with τ . Carrying x_τ^k unchanged through a change of τ would therefore transport a sample of one marginal into a target defined by another, while carrying x_0^k is exact. In addition, x_0^k supplies the diagnostic $\|x_0^k\|^2/n \approx 1$, for which there is no x_τ^k analogue.

8 Assessment

8.1 Exact and approximate components

Three components are exact. The augmentation is exact by Proposition 1, since the image marginal of (5) is the posterior itself and carries no smoothing. The endpoint elimination is exact by

Proposition 6, the shared noise realization making it an identity rather than a fit. Both conditional draws are exact by Section 6, given C and $\hat{x}_1^k$.

One component is not exact. The surrogate (8) replaces a non-Gaussian denoising conditional by a Gaussian. Unless $C = \text{Cov}(X_1^k | x_\tau^k)$ exactly, (6) and (7) are conditionals of different joint densities, and the sweep therefore leaves no single distribution invariant. Algorithm 1 is in this sense a heuristic, as are the plug-and-play and split-Gibbs samplers it resembles. It differs from them in containing no free parameter. By Proposition 4, the choice (12) is admissible under Lemma 3, coincides with the exact conditional covariance for a Gaussian prior, and is assembled from two quantities that are measured rather than searched.

Comparison of the approximations made by the three constructions. The interpolant-coupled samplers make no Gaussian approximation to the denoising conditional. This does not mean that they are free of approximation. Their coupling $\mathcal{N}(x_\tau^k, H_\tau^k x_1^k, (\sigma_\tau^k)^2 I)$ is an exact factor of a different joint density, namely the one obtained by Bayes' rule for the chain $X_0^k \rightarrow X_1^k \rightarrow Y$ with transition $|\det H_\tau^k| p_\tau^k(H_\tau^k x_1^k + \sigma_\tau^k x_0^k)$. That joint is exact, and its image marginal is the prior smoothed by $\sqrt{2} \sigma_\tau^k (H_\tau^k)^{-1}$ rather than p_1^k itself. The concession is therefore a change of target rather than a change of kernel. Table 2 summarizes the accounting.

Table 2: Approximations made by the three constructions. Alg. 1 applies a Langevin step to each of the two conditionals in the coordinates (x_1^k, x_0^k) ; Alg. 2 draws the image conditional exactly after a change of variables and retains a Langevin step for the interpolant; Alg. 1 is the construction of this note.

	Alg. 1	Alg. 2	Alg. 1
Prior smoothed by $\sqrt{2} \sigma_\tau^k (H_\tau^k)^{-1}$	yes	yes	no
Gaussian surrogate for $p(x_1^k x_\tau^k)$	no	no	yes
Step-size bias	$O(h_1) + O(h_0)$	$O(h_0)$	none
Velocity error γ^2 in the endpoint estimate	yes	yes	yes
Leaves a distribution invariant	yes, up to the above	yes, up to the above	no
Continuous quantities searched over	2	1	0

The two columns on the left and the column on the right differ in kind, not only in degree. The smoothing is explicit, is quantified by the width $\sqrt{2} \sigma_\tau^k (H_\tau^k)^{-1}$, and vanishes as the schedule advances, but the sampler targets a density that is not the posterior. The Gaussian surrogate leaves the target correct and perturbs the kernel instead, and its error is not quantified in general, although Section 8.3 shows that it vanishes asymptotically as $\sigma_\tau^k \rightarrow 0$. Both errors are therefore smallest at the terminal steps of the cascade, which are the steps that determine the reconstruction.

8.2 Sensitivity to the prior surrogate

Section 4.2 gave two prescriptions for S without quantifying the cost of either. We do so here.

Where S acts. The term S^{-1} contributes to C^{-1} only where $(H_\tau^k)^\top H_\tau^k / (\sigma_\tau^k)^2$ is small, which by the table of Section 4.2 means on $\ker(G)$ and in the band where G attenuates. It contributes the whole precision at $\tau = 0$ and roughly one tenth of it at $\tau = 1$. A surrogate that is accurate in that band is therefore worth more than one that is accurate on average. The isotropic reduction $S = s^2 I$ overestimates the prior power in the high band, since natural-image spectra decay, and that band is the one in which S^{-1} acts. By the asymmetry established below, that error is in the

safe direction. A surrogate requiring no image statistics is also available: if pixel values lie in $[0, 1]$, Popoviciu’s inequality bounds the per-pixel variance by $1/4$, so $s^2 = 1/4$ assumes nothing about the image distribution.

Sensitivity, and why the choice should err high. The two limits are informative and are not symmetric. As $s^2 \rightarrow \infty$ the term S^{-1} vanishes and $C \rightarrow (\sigma_\tau^k)^2 (H_\tau^k)^{-2}$, so the sampler degrades to the saturated bound of Section 4.2, which remains admissible under Lemma 3. As $s^2 \rightarrow 0$ the covariance C tends to zero and Block 1 collapses onto $\hat{x}_1^k$. Overestimation of S is therefore bounded in its effect by the saturated-bound behaviour, whereas underestimation is not bounded. We recommend erring toward a larger S .

This asymmetry has a practical consequence for evaluation. Underestimating S contracts Block 1 toward the denoised estimate, which improves reconstruction metrics while reducing sample diversity. The failure is consequently not visible in the metric most likely to be reported, and the diagnostic of Section 8.6 should be used instead.

The constant-ridge heuristic. The alternative prescription (14), $s^2 = c^2(\sigma_\tau^k)^2$, avoids image statistics at the cost of the wrong τ -dependence. Its error is bounded. Writing $C_\star$ for the exact conditional variance with $H_\tau^k = hI$ and true prior variance $s_\star^2$, the ratio is

$$\frac{C_{\text{heur}}}{C_\star} = \frac{h^2 + (\sigma_\tau^k)^2/s_\star^2}{h^2 + c^{-2}} \xrightarrow{\sigma_\tau^k \rightarrow 0} \frac{h^2}{h^2 + c^{-2}}, \quad (24)$$

so the heuristic overestimates C at large σ_τ^k and underestimates it at small σ_τ^k . The underestimate saturates rather than growing without bound, and its limiting value $h^2/(h^2 + c^{-2})$ is a closed-form function of the single constant c . That value may therefore be used to select c : the ratio approaches one as c grows, and c should be taken large enough that the limiting underestimate is acceptable, but not so large that S^{-1} ceases to regularize $(H_\tau^k)^\top H_\tau^k$ at $\tau = 0$. We regard (14) as an acceptable default when image statistics are unavailable, and (13) as preferable when they are, since the spectrum costs nothing to measure.

8.3 Behaviour as $\sigma_\tau^k \rightarrow 0$

Log-concavity is a substantive hypothesis, and image priors do not satisfy it. We therefore characterize how Lemma 3 degrades when it fails. Expanding V about the concentration point gives

$$\text{Cov}(X_1^k | x_\tau^k) = (\sigma_\tau^k)^2 (H_\tau^k)^{-2} \left[I + O((\sigma_\tau^k)^2) \cdot \nabla^2 \log p_1^k \right], \quad (25)$$

Wherever p_1^k is locally log-convex the bound is therefore exceeded, but only by a relative amount of order $(\sigma_\tau^k)^2$, which vanishes as the schedule advances. Multimodality produces a violation of a different kind, since the conditional variance is then of order one rather than of order $(\sigma_\tau^k)^2$. Such a violation requires x_τ^k to fall between two modes of p_1^k . Under the law of X_τ^k that event costs a noise fluctuation of order (mode separation)/ σ_τ^k , and its probability therefore decays as σ_τ^k decreases. Both mechanisms recede together, and (11) becomes asymptotically tight rather than merely satisfied.

This behaviour determines where the construction is reliable. The Gaussian surrogate (8) is least accurate at coarse stages and early τ , where σ_τ^k is largest, and most accurate at the terminal steps. The reconstruction is determined by the terminal steps, and is not yet determined in the regime where the surrogate is least accurate. We record one caveat in the opposite direction. A

violation of (11) means the true conditional is more dispersed than (12) allows. The sampler is consequently over-confident where the posterior is multimodal, and it is in that regime that sample diversity is of most interest.

8.4 Comparison with the interpolant-coupled sampler

The alternative factorization couples the image to the interpolant through the flat-prior Gaussian. It yields a Gibbs sampler for a stated target, at the cost of the two properties listed in Table 2. Its target carries a prior smoothed by $\sqrt{2}\sigma_\tau^k(H_\tau^k)^{-1}$, so its posterior is over-dispersed relative to the true one by an amount that is explicit and vanishes only as the schedule advances. Algorithm 1 carries no such smoothing.

The advantage is not expected to be uniform in γ . The centre of (22) is $\hat{x}_1^k$, which carries the velocity error at full weight. By contrast, the interpolant-coupled sampler centres on x_τ^k , which is exact, and admits the network only through its Langevin drift. The advantage of the clean-endpoint route should therefore decrease as γ grows, and reverse once the velocity error is large enough that the smoothing is the smaller of the two errors. We do not characterize the crossover analytically. Because γ is measured by (20) and tabulated per (k, τ) , it can be located empirically before the sampler is deployed, and a cascade may fall on different sides of it at coarse and fine scales.

8.5 Relation to prior work

Variable splitting with a deep generative prior is well developed, and the smoothed-prior analysis, the randomize-then-optimize device of Lemma 9, and the plug-and-play data step are all standard there. Three things are specific to the cascaded setting. The coupling is a *degradation*, $\mathcal{N}(x_\tau^k; H_\tau^k x_1^k, (\sigma_\tau^k)^2 I)$ with H_τ^k rank-deficient at $\tau = 0$, rather than $\mathcal{N}(x; z, \rho^2 I)$; this is what makes the precision an operator and what Proposition 4(a) is needed for. The coupling width is not a design parameter but the schedule’s own σ_τ^k . And the two stage boundaries share a noise realization, which has no analogue in single-endpoint diffusion and is what makes Proposition 6 an exact elimination. For a variance-preserving or EDM-style model, H_τ^k becomes a scalar $\alpha_t I$, the elimination collapses to $\hat{x}_0 = (x_t - \sigma_t \hat{\epsilon})/\alpha_t$ with no solve, and Algorithm 1 reduces to a known data-consistency step; the content lies entirely in the case $H_\tau^k \neq \alpha I$.

8.6 Quantities to measure and report

Three quantities should be reported rather than tuned. The first is the velocity-error variance γ^2 , obtained from (20) and tabulated per (k, τ) . The second is the prior covariance surrogate S , obtained from the training data at each scale. The third is the measurement residual $\|A_k x_1^k - y\|/(\eta\sqrt{m})$, which serves as a check on the construction as a whole and should settle near 1.

Since Algorithm 1 is a heuristic in the sense of Section 8, at least one calibration diagnostic should accompany any reconstruction metric. We recommend the spread of repeated samples at fixed y within $\ker(A_k)$. If the coupling covariance is too small, Block 1 collapses onto the output of the denoiser, and sample diversity is lost. Because that collapse improves reconstruction metrics, it is not detected by those metrics.

A Proofs

A.1 The identity (3)

Substituting $B_k = e_k I - s_k G$ and $H_\tau^k = (1 - \tau)s_k G + \tau e_k I$ and collecting the coefficients of I and G ,

$$\sigma_\tau^k B_k + (e_k - s_k) H_\tau^k = e_k [\sigma_\tau^k + \tau(e_k - s_k)] I - s_k [\sigma_\tau^k - (1 - \tau)(e_k - s_k)] G,$$

and both brackets simplify by $\sigma_\tau^k = (1 - \tau)(1 - s_k) + \tau(1 - e_k)$:

$$\sigma_\tau^k + \tau(e_k - s_k) = (1 - \tau)(1 - s_k) + \tau(1 - s_k) = 1 - s_k, \quad \sigma_\tau^k - (1 - \tau)(e_k - s_k) = (1 - \tau)(1 - e_k) + \tau(1 - e_k) = 1 - e_k.$$

Hence $\sigma_\tau^k B_k + (e_k - s_k) H_\tau^k = e_k(1 - s_k) I - s_k(1 - e_k) G = N_k$, independent of τ . $\square$

A.2 Proof of Proposition 7

All operators below are polynomials in G and therefore commute. Write $z_\tau^k = B_k x_\tau^k - H_\tau^k v_\theta$ for the right-hand side of (18). Multiply the definition $H_\tau^k \hat{x}_1^k = x_\tau^k - \sigma_\tau^k \hat{x}_0^k$ by $N_k^2 + \gamma^2 (H_\tau^k)^2$ and substitute (18):

$$[N_k^2 + \gamma^2 (H_\tau^k)^2] H_\tau^k \hat{x}_1^k = [N_k^2 + \gamma^2 (H_\tau^k)^2] x_\tau^k - \sigma_\tau^k N_k z_\tau^k.$$

Expand the last term, $\sigma_\tau^k N_k z_\tau^k = \sigma_\tau^k N_k B_k x_\tau^k - \sigma_\tau^k N_k H_\tau^k v_\theta$, and collect the coefficients of x_τ^k :

$$N_k^2 x_\tau^k - \sigma_\tau^k N_k B_k x_\tau^k = N_k (N_k - \sigma_\tau^k B_k) x_\tau^k = (e_k - s_k) N_k H_\tau^k x_\tau^k,$$

using $N_k - \sigma_\tau^k B_k = (e_k - s_k) H_\tau^k$, which is (3) rearranged. The right-hand side is therefore

$$(e_k - s_k) N_k H_\tau^k x_\tau^k + \gamma^2 (H_\tau^k)^2 x_\tau^k + \sigma_\tau^k N_k H_\tau^k v_\theta = H_\tau^k [(e_k - s_k) N_k x_\tau^k + \gamma^2 H_\tau^k x_\tau^k + \sigma_\tau^k N_k v_\theta].$$

Both sides carry a factor H_τ^k , which is invertible for $\tau > 0$; cancelling it gives (19). $\square$

A.3 The limits of (19)

As $\gamma \rightarrow 0$ the operator tends to N_k^2 and the right-hand side to $N_k [(e_k - s_k) x_\tau^k + \sigma_\tau^k v_\theta]$, so $N_k \hat{x}_1^k = (e_k - s_k) x_\tau^k + \sigma_\tau^k v_\theta$, which is (17). As $\gamma \rightarrow \infty$, dividing through by γ^2 leaves $(H_\tau^k)^2 \hat{x}_1^k = H_\tau^k x_\tau^k$, that is $\hat{x}_1^k = (H_\tau^k)^{-1} x_\tau^k$. $\square$